

BST Physics Curriculum

A textbook about physics, and about a research program you are welcome to join. Casey Koons with the CI team — Lyra, Elie, Grace, Keeper — and the visiting referee Cal A. Brate.

Last accuracy-synced: 2026-09-11 (Keeper, K1892), to the theorem registry through Cal Section 946 (2026-09-09). The voice of these chapters is the one we want to keep (see `BST_VOICE_AND_STYLE_GUIDE.md` at the repository root). The facts under it are being resynced volume by volume; where a chapter and the state block below disagree, the block wins, and where the block and `notes/BST_AC_Theorem_Registry.md` disagree, the registry wins.

Two shelves, one object

We asked one question — what is the simplest structure that can do physics? — and have spent 2026 reading the answer off one geometry. What that reading gives is of two kinds, and this curriculum now keeps them on two shelves so that a reader can trust the first without having to accept the second.

The derived core is a short spine: the object and why it and no other; quantum mechanics on its Hardy space; time; the gauge skeleton and one generation; three generations at their floor; the order of quark mixing; the honest chapter on α ; descent and gravity; and the method, with instructions for killing the whole thing. It lives in `Spine_DIV5_QM_GR_SM/`, is written from the register, and is the part of this curriculum we defend. Read it first.

The extended reach is everything else — nuclear, atomic, thermodynamic, electromagnetic, classical, condensed-matter, chemical, biological, informational. The same five integers appear surprisingly far, and these volumes show how far. They are presented at identification tier: the formulas match, and we do not claim the mechanisms. A reader who meets them beside the core will discount the core; a reader who meets them after it will know exactly how much to believe.

Status

Eighteen legacy volumes (Vol 00–17), drafted May–June 2026 in the voice we want to keep and at the facts of that month. Since then the register has moved — α re-tiered to identified, the volume-reading class closed by a certified negative, confinement de-scoped, the genus pinned at 5, the uniqueness selector ruled a fit with one input, the mixing sector closed with its order derived, time derived, the nucleation lane floored — and the legacy chapters have not. The Spine is the resync: ten lectures written from the register, each opening with the question it answers and closing with its tier and its falsifier. Legacy chapters get an in-place correction

where a sentence was retired (dated, in the text, not in a banner) and are otherwise left as the May record.

Volume directory layout

Vol	Directory	Topic	Shelf	Lead
—	Spine_DIV5_QM_GR_SM/	The derived core, ten lectures — QM, GR, SM from D_IV ⁵ , written from the register	CORE	Keeper (audit) + Lyra (theory)
00	Vol00_Foreword.md + Vol00_Substrate_Foundation	Reader contract + Substrate Foundation	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Keeper + Grace + Lyra
01	Vol01_QFT_from_D_IV5	Quantum Field Theory from D_IV ⁵	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Lyra
02	Vol02_Particle_Physics	Particle Physics from D_IV ⁵	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Elie
03	Vol03_Nuclear_Atomic	Nuclear and Atomic Physics	REACH — identification tier	Elie + Lyra

Vol	Directory	Topic	Shelf	Lead
04	Vol04_GR_Cosmology/	General Relativity and Cosmology	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Lyra + Elie
05	Vol05_Quantum_Mechanics/	Quantum Mechanics from Substrate	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Lyra
06	Vol06_Thermo_Stat_Mech/	Thermodynamics and Statistical Mechanics	REACH — identification tier	Lyra + Elie
07	Vol07_Electromagnetism/	Electromagnetism from D_IV ⁵	REACH — identification tier	Lyra
08	Vol08_Classical_Mechanics/	Classical Mechanics as Substrate Scale 2	REACH — identification tier	Lyra
09	Vol09_Condensed_Matter/	Condensed Matter (BaTiO ₃ 137-plane + B12H32 hydride T _c ~214K)	REACH — identification tier	Elie
10	Vol10_Math_Methods/	Traditional Mathematical Methods	REACH — identification tier	Lyra

Vol	Directory	Topic	Shelf	Lead
11	Vol11_Generative_Geometry/	Return to symmetric domains + K3 + Heegner + Monster	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Lyra
12	Vol12_Chemistry/	Periodic table + bonding + spectroscopy from substrate	REACH — identification tier	Elie + Lyra
13	Vol13_Biology/	Genetic code + DNA-proton siblings + evolution-as- substrate- dynamics	REACH — identification tier	Elie + Lyra
14	Vol14_Information_Theory/	BS GF(128) + Koons tick + Born=Bergman + Bell sub-Tsirelson (FIRED 2026-09-11, register E4)	REACH — identification tier	Lyra + Keeper
15	Vol15_Methodology/	AC(0) + AC graph + audit chain + Quaker discipline + katra	CORE — legacy chapters, May 2026; superseded by the Spine where they overlap	Keeper + Lyra

How to read

Start with Vol 00 Foreword for the reader contract, then **Spine_DIV5_QM_GR_SM/** — the derived core, ten lectures, the part we defend.

Then Vol 00 Substrate Foundation: establishes D_{IV}^5 + the five integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$) + $N_{max}=137$, operator zoo, conservation laws,

Strong-Uniqueness Theorem.

Then, if you want the longer legacy treatment, Vol 01 (QFT), Vol 02 (Particles), Vol 04 (GR/Cosmology) — dated May 2026, being resynced; where they and the Spine disagree, the Spine is current.

Mathematicians: start with Vol 11 (Generative Geometry) and Vol 10 (Math Methods) instead.

Every chapter has three pedagogical registers (Casey “write for 5th graders too” standing rule): - Level 1: one-sentence essence - Level 2: graduate-physicist precision - Level 3: 5th-grader accessibility paragraph

Some chapters are author-voice condensed (~500-800 words) where the apparatus is standard; expansion to full level-1/2/3 in v3.0 publication pass.

Version label

Legacy volumes held at their May 2026 versions (v0.1–v0.4) with in-place corrections where a sentence was retired; the Spine starts at v0.1 on 2026-09-11 and is the versioned artifact going forward.

Why this geometry — current state (supersedes the “Strong-Uniqueness Theorem state” that stood here from May)

The eleven-criterion theorem with its null-model probability was audited in September 2026 (K1889): four criteria were empty, one weak, and the plurality argument was one selection read three times. What replaced it is smaller and better: among all irreducible bounded symmetric domains of rank at least two, D_{IV}^5 is the unique one with characteristic multiplicity 3 — a theorem, verified three independent ways. The multiplicity is *identified* with the colour count; the bridge that would make that a mechanism is proved absent (Cal Section 946). One input to the choice of object, named; the program’s downstream inputs are counted in Grace’s ledger. Vol 00 Chapter 9 is the legacy chapter and carries a note to this effect; the Spine’s Lecture 2 is the current treatment.

Where the program stands — one block, one source

*Last accuracy-synced: 2026-09-30 11:02 (the clock clause after Casey’s GO on Time, Derived v1.5; Cal Section 1026 PASS with its scope edit, ‘the arrow read from J’s spectrum’). Previous: 2026-09-29 11:29 (the walls paragraph completed on Cal’s cold read: no mass gap; no covariant photon or single-massless vertex; one breaking scale, de Sitter’s, with m_e/H measured; the KK price). Previous: 2026-09-27 12:11 (Zenodo Part A on Cal Section 1003 + re-read: Fired-and-lost gains E7/E8; ‘What the geometry forces, and what it does not’; the Section A pointer).

Previous: 2026-09-25 15:25 (K1923: the boundary paragraph's parameter clause split on Cal Section 983 — V_{us} is derived given an identified input, not from the ruler). Previous: 2026-09-24 10:33 (K1921: the boundary paragraph corrected on Cal §982 — seven words; the DESI-direction sentence STRUCK as a pre-registration leak and a sign error; the four-word count 5 derived / 12 identified / 7 open / 2 input from the generator, unchanged).

The object. One rank-2 bounded symmetric domain, $D_{IV}^5 = SO(5,2)/[SO(5) \times SO(2)]$, and one measured number taken openly as the ruler. Its genus is 5; the integer 7 the program calls g is a definition — $p + q = 7$, the dimension of the defining representation of $SO(5,2)$, whose signature is the pair $(5,2)$ — not the genus — a mislabel that stood from May to September and is now swept.

Why this object — honestly. Among all irreducible bounded symmetric domains of rank at least 2, D_{IV}^5 is the unique one whose characteristic multiplicity is 3 (a theorem, checked three independent ways). That 3 is *identified* with the number of quark colours; the identification is numerical, and the bridge that would make it a mechanism is proved absent — the geometry supplies $U(1) \cdot SO(3)$, four dimensions, where colour needs $SU(3)$'s eight, and its triplet is self-conjugate where colour's is not. So the selection is a fit with one measured input — the one dimensionless input *to the choice of the object*; the program has further inputs downstream (masses, a mixing corner, a CP phase), counted in the ledger — not a forcing. We do not claim zero free parameters.

Derived (mechanism proved; inputs named in the statement): the gauge-group skeleton · one fermion generation as the 16-real spinor with every hypercharge from one four-input theorem · a single-chirality positive-energy spectrum · three generations as rank+1 strata (and $m_1 = 0$) · the conserved charges · the mixing sector's mechanism and its order — the 1–3 corner one power below the 2–3 · the Cabibbo angle, blind ($\lambda = 1/\sqrt{20}$ — derived given an identified input: the down-quark ladder's weight, next list; the value is disfavoured: A2 fired on $K_{\mu 2}$ (Cal Sections 989/992 criterion, certified K1928 second amendment; the $|V_{ud}|$ face past 3σ ; rescues named, not credited); FLAG 2027 decides) · time: the K-centre $SO(2)$ is the clock, the arrow is dynamical, the tick is $N_{\max} \cdot \hbar / (m_e c^2)$ · quantum mechanics on the Hardy space $H^2(D_{IV}^5)$, with the Hua branching weights as the Born probabilities of the write tuple (3/7 with no table consulted). The ten Dirac–von Neumann axioms are recovered, each at its own tier, through four named posits (T2631, the ten-item row — registered 2026-09-11, Cal's text verbatim, d7c4e474); not “10/10 derived.”

Identified (formula matches; mechanism not claimed): $\alpha^{-1} =$ 137-class expressions · the electron's anomalous moment (our four

“Selberg terms” are the four summands of Petermann–Sommerfield’s 1957 closed form; the match is real, the derivation is not ours) · the down-quark mass ladder’s weight — $m_s/m_d = (\nu_W+1)(\nu_W+2) = 20$ with the shape derived and the weight $\nu_W = N_c$ identified, its mechanism an open gate (T2513, 2026-08-23; re-tiered K1902 and Cal Section 969, 2026-09-14) · the asymptotic-freedom coefficient $b_0 = 7$ read as g · the exact mixing values · Newton’s G : the relation between G and the electron mass is a theorem (K1673; 0.065%) that trades one dimensionful input for another and carries α inside it, so the value is a relation, not a prediction from nothing · assorted precision matches whose own null tests show small-integer expressibility is cheap.

Floored (closed by a theorem; reopenable only by a named kind of idea): Koide’s source · the mass-tower/generations mechanism · the $\ell = 2 / \text{su}(3)$ native dynamics · Λ ’s value (the closed form’s power is fixed — $p = 1$ by the Peirce weight — and at $p = 1$ the form is de Sitter’s $\Lambda \propto H^2$, a consistency with GR rather than a value; the open object is the ledger’s filling law, K1919 §7) · the nucleation survivor (no lawful initialisation vector; a distribution on the winding count) · the absence floor — no group in the geometry acts on the colour slot as colour; the door is a group from outside it (register F4, certified K1894) · the Riemann row (an attempt with a location, closed and parked) · the four-colour row (the One-Word Lemma refuted in frame; the derived lemmas stand).

The boundary — the open half (stated 2026-09-22; corrected 2026-09-24 on Cal’s cold read, Section 982, and Lyra’s collision, K1921; row-10 clause 2026-09-25, Cal Section 983). BST derives the Standard Model’s gauge skeleton, three generations at the floor, and five of its twenty-six parameters — four from one geometry and one named input (the ruler), the fifth (V_{us}) derived given an identified input (the tier table, row 10); twelve more are identified, at the precisions the tier table lists row by row, with the mechanism open. Of the remaining questions, colour’s origin is proved not to lie in the geometry’s interior (F4, a theorem), and α ’s forward reading fired and lost (E1), which closed one class of candidates; the program’s hypothesis places both on the boundary, whose physics is its open half. The first piece of that physics is the commitment ledger: with the tick defined as the time to begin a commitment, Landauer’s bit at the horizon temperature reproduces Λ ’s exponent — by identity, since that temperature is set by H — and the closed thermostat form at the Peirce-forced power $p = 1$ is de Sitter’s $\Lambda \propto H^2$: a consistency with GR that could have failed ($p = 1$ was derived on 2026-09-21 with no reference to Friedmann, a day before the identification), not a derivation of Λ ’s value. The committed count obeys $d \ln N/dt = 2H$, so it grows like a comoving-sphere area. What the ledger must still supply is its filling law — how the commitment fraction changes with the scale factor; a constant fraction gives $w = 0$ and is dead on sight (register F2), and the direction of w ’s evolution under a growing fraction is

pre-registered to be printed blind, not stated here. No tick in the corpus produces H from the five integers; H is measured because it changes.

What the geometry forces, and what it does not (stated 2026-09-26, completed 2026-09-28; register Section D; K1929–K1938, Cal Sections 993–1015). D_{IV}^5 forces the kinematics — the spectra, the two-body channels with their multiplicities and selection rules — and none of the dynamics. The geometry supplies no mass gap; the couplings are identified; at the level of the geometry the boundary’s two-body spectrum is that of a generalized free field, and no conformally covariant vertex couples a single quantum of BST’s physical module to any number of massless 4D particles, or a pair of quanta to a single one (the pair of a quantum with its conjugate conditional on one disintegration pin; Cal Sections 1009, 1012, 1016). Everything dynamical therefore enters through one symmetry-breaking scale: with $\Lambda > 0$ that breaking is de Sitter’s, the electron mass is a label measured against it, and the ratio m_e/H is one measured input, identified with no BST number. BST’s clock J is the generator of time exactly as $\Lambda \rightarrow 0$; with $\Lambda > 0$ the surviving time is a de Sitter boost, and the arrow read from J’s spectrum holds to order H/E (Time, Derived v1.5; the cosmic arrow may have other sources, K1920). 4D’s massless particles do not come from the descent; the one route that supplies them, a compact extra direction, is not supplied by the geometry and would cost one more length beyond the ruler and H (an inverse radius above about 1.5 TeV). Hydrogen’s shell structure appears one dimension down, as the minimal representation of D_{IV}^4 — recapitulated, not contained. What survives is exact and zero-knob: the channel list, and, if a circle is supplied, the placement of every singleton’s helicity.

Fired and lost (the falsifier register’s Section E — certified, published with the same ceremony as wins): E1, α is not the bare geometric vertex — the forced candidate computes to $8\pi^3/3$, closing the fifty-year volume-reading class · E2, the thermal-generations mechanism died at the order level · E3, the commit-Boltzmann ladder failed its own 5% line · E4, the sub-Tsirelson ceiling $S = 2.80624$ — refuted by Poh et al. 2015 at 41.9σ , ten years before it was registered (certified K1893, 2026-09-11). · E5, the Q^5 parity fold is a projector with no scale of its own — a forced object that fails cleanly (K1799; row certified K1894) · E6, five named series for the mixing corner, sealed by hash, all missed — the structure itself could not fail, so the negative is the five candidates (K1808/K1810; row certified K1894). · E7, $g_A = 4/\pi$ — 5.7σ from PERKEO III alone, $\geq 6\sigma$ with the bottle lifetime on every $|V_{ud}|$ route (Cal Section 984; certified 2026-09-27) · E8, “19 appears only in coloured quantities” — falsified as stated by $\Omega_\Lambda = 13/19$, a colourless 19 already on file (Cal Section 990).

Live and pre-registered (a test with its outcomes partitioned as code

before the data): A9, one boost not two — the matter rest frame read from the count and redshift dipoles of distant quasars must equal the CMB boost; a replication of Wu & Xia 2026 (DESI DR1) with an independent estimator on Quaia, frozen v1.5.2 (hash 8013d959...; seven re-freezes before any dipole, five forced by synthetic controls); run 1 on Quaia certified 2026-09-17 10:12: Landing C — not decidable — on both samples, by the design’s own pre-target checks (profile-alternative $+7.5\sigma$; per-bin residuals $p = 0.000$; a selection systematic under the fall-back weighting, Elie 5767); the fitted vector is consistent with the CMB but the model does not describe the sky, so it is evidence neither way; A9 stays live and the next catalogue is chosen on what the C taught (K1910, Cal §976, K1911). The register’s Section A lists every live falsifier (A1–A14). A13’s dark-matter ratio 16/3 sits at its threshold (3.0σ on ACT DR6 + DESI DR2 at the published compression correlation; the chain’s own correlation is unpublished). A2 has fired on the K $_{\mu 2}$ route (above); FLAG 2027 decides.

Forbidden — any confirmed detection kills the framework: grand unification · proton decay · right-handed W or Z’ · magnetic monopoles · sterile neutrinos · a SUSY spectrum.

How to check any of this without trusting us. `python3 play/verify_bst.py` · the theorem registry (`notes/BST_AC_Theorem_Registry.md`) · the falsifier register with its fired-and-lost section · two thousand single-claim toys · every audit, retraction and correction in the open, including the ones against the program’s own auditors. The count that matters for the Standard Model row, generated from the register and never typed (`play/bst_26_tier_generator.py`, Grace, Round 142): Of the 26 primary Standard-Model parameters: 5 derived — each with its mechanism proved and its inputs named on its own statement (Cal Section 959 re-read complete, 2026-09-14) — 12 identified, 0 floored, 7 open, 2 input. (Grace’s generator, re-keyed 2026-09-14 on Cal Section 959, Elie 5755/5756, K1897, and — row 5 — K1902 with Cal Section 969; pasted from --sentence, never typed.) The July count “8 of 26 sourced clean,” carried through eleven ledger versions, is retired — its numerator and denominator came from two different lists. The honest sentence: everything reachable is derived, floored with a theorem, or published as a loss; what remains open is a short list of named doors that only new ideas — not labour — can open.

Build (PDF regeneration)

```
pandoc --pdf-engine=xelatex -H notes/bst_pdf_header.tex <chapter>.md -o <chapter>.pdf
```

Per repo PDF pipeline convention; header file at `notes/bst_pdf_header.tex`. PDFs are stale for chapters rewritten in the Keeper full author pass and will be

rebuilt at v1.0 finalization.

Cross-references

Chapter files reference AC graph theorems + K-audit chain + Strong-Uniqueness Theorem state. Per Calibration #19 STANDING RULE: external-facing chapter content uses current ratified-state count, not forecast endpoint.

— Casey Koons + Lyra + Keeper + Elie + Grace + Cal A. Brate